

中山大學
SUN YAT-SEN UNIVERSITY

Project

Data Structures and Algorithms

Zizhen Zhang, School of Computer Science and Engineering, Sun Yat-sen University

zhangzizhen@gmail.com

Project Assignment

- Complete a small text searching engine using hash tables.
- Basic requirements:
  - Excerpt an essay (1KB<=essay size<=1MB) from anywhere and save it as “text.txt”.
  - Write a program for the text searching (English & Chinese).
  - The program receives several queries. Each query contains a keyword.
  - You program should output all the sentences containing this keyword. (A sentence is ended up with a period “.”)
  - Test different queries (at least 10 queries) by yourselves.
  - Try to use different hashing functions and/or collision resolving methods.
- Extend your work as you like.
- Discuss the advantage and deficiency of your work.

- Submit your assignment (inputs/outputs, codes and report >10p.) in *.zip* or *.rar*.
- Deadline: Dec. 15, 2024.

Project Assignment

- Example
  - "text.txt":*In the beginning God created the heaven and the earth. And the earth was without form, and void; and darkness was upon the face of the deep. And the Spirit of God moved upon the face of the waters. And God said, Let there be light: and there was light.*
  - Query 1: *God*
  - Output1:

    In the beginning God created the heaven and the earth.

    And the Spirit of God moved upon the face of the waters.

    And God said, Let there be light: and there was light.
  - Query 2: *earth*

- Output 2:

  And the earth was without form, and void; and darkness was upon the face of the deep.

Thank you!



